

Problem Statement 3:

Smart AI-Enabled Rapid Feed and Silage Quality Testing System for Dairy Farmers

Background:

Animal nutrition directly affects milk production, animal health, reproductive performance, and dairy profitability. Dairy farmers often face challenges due to poor-quality cattle feed, adulterated feed ingredients, fungal contamination, toxin presence, and low-quality silage.

Conventional feed testing laboratories are expensive and inaccessible for many rural farmers. There is a need for rapid, portable, affordable, and digitally enabled feed quality assessment systems.

Emerging technologies such as AI, IoT, spectroscopy, computer vision, and biosensors can help create real-time feed testing and advisory systems for dairy farmers.

Description:

Participants are required to develop a rapid digital testing solution capable of:

- Assessing nutritional quality of cattle feed and silage;
- Detecting adulteration and contamination;
- Providing instant farmer advisories and feed recommendations;
- Monitoring feed storage and silage conditions.

The solution may include:

- Portable testing devices;
- Smartphone-enabled feed analysis;
- AI-powered nutritional prediction;
- Cloud dashboards;
- QR-based authenticity systems.

The system may detect:

- Crude protein
- Moisture
- Fiber
- Energy value
- Mineral deficiencies
- Urea adulteration
- Sand/silica contamination
- Aflatoxins and mycotoxins
- Fungal contamination

Silage monitoring may include:

- pH
- Fermentation quality
- Moisture
- Spoilage indicators
- Mould growth

Expected Solution:

The expected solution should:

- Provide testing results within minutes;
- Be low-cost and portable;
- Support multilingual farmer interfaces;
- Work offline in rural areas;

- Generate nutritional and storage advisories;
- Enable cloud-based monitoring and traceability.

Expected technologies:

- AI/ML
- IoT sensors
- NIR spectroscopy
- Mobile applications
- Computer vision
- Cloud analytics
- Predictive advisory systems

Dummy Data Set

Feed Sample ID	Feed Type	Moisture (%)	Protein (%)	Fiber (%)	Aflatoxin (ppb)	pH	Adulteration Detected	Quality Status
F001	Cattle Feed Pellet	10	21	14	5	NA	None	Good
F002	Silage	68	9	24	12	4.1	Mould Presence	Moderate
F003	Mineral Mixture	3	NA	NA	0	NA	Excess Salt	Poor
F004	Feed Mash	12	18	16	20	NA	Sand Contamination	Unsafe
F005	Silage	72	8	28	8	5.8	Spoilage Detected	Poor